

Debiased/Double Machine Learning (DML)

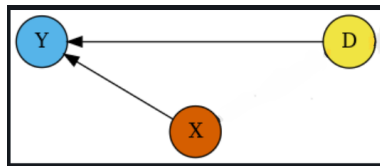
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Initial Situation

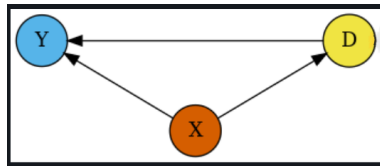
$$E[D|X] = 0$$

Gold standard of every inference study:



The situation we usually face:

$$E[D|X] \neq 0$$



The Partially Linear Regression

$$Y = D\theta_0 + g_0(X) + U, \quad \mathbb{E}[U \mid X, D] = 0$$

$$D = m_0(X) + V, \quad \mathbb{E}[V \mid X] = 0$$

Why this setup?

- We can suppose $g_0(\cdot)$ is **as complicated as we want**.
- It's straightforward in the application of the DML.

The Naive Approach

The standard formula to estimate a linear predictor is:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Applied to our case, the naive approach becomes:

$$\hat{\theta}_0 = \left(\frac{1}{n} \sum_{i \in I} D_i^2 \right)^{-1} \frac{1}{n} \sum_{i \in I} D_i (Y_i - \hat{g}_0(X_i))$$

Regularization Bias

The Scaled Bias Formula:

$$\begin{aligned}\sqrt{n}(\hat{\theta}_0 - \theta_0) = & \underbrace{\left(\frac{1}{n} \sum_{i \in I} D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i \in I} D_i U_i}_{:=a} \\ & + \underbrace{\left(\frac{1}{n} \sum_{i \in I} D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i \in I} D_i (g_0(X_i) - \hat{g}_0(X_i))}_{:=b}\end{aligned}$$

Term a

Converges to zero
under mild conditions.

Term b (Regularization Bias)

$$b = (E D_i^2)^{-1} \frac{1}{\sqrt{n}} \sum_{i \in I} m_0(X_i) (g_0(X_i) - \hat{g}_0(X_i)) + o_P(1)$$

Partialling-out

$$V = D - m_0(X) \implies \hat{V} = D - \hat{m}_0(X)$$

The Orthogonalized Estimator:

$$\check{\theta}_0 = \left(\frac{1}{n} \sum_{i \in I} \hat{V}_i D_i \right)^{-1} \frac{1}{n} \sum_{i \in I} \hat{V}_i (Y_i - \hat{g}_0(X_i))$$

New Scaled Error

$$\sqrt{n}(\check{\theta}_0 - \theta_0) = a^* + b^* + c^*$$

$$a^* = (\mathbb{E} V^2)^{-1} \frac{1}{\sqrt{n}} \sum_{i \in I} V_i U_i \rightsquigarrow \mathcal{N}(0, \Sigma)$$

$$b^* = (\mathbb{E} V^2)^{-1} \frac{1}{\sqrt{n}} \sum_{i \in I} (\hat{m}_0(X_i) - m_0(X_i))(\hat{g}_0(X_i) - g_0(X_i))$$

Cross-fitting

$$c^* = \frac{1}{\sqrt{n}} \sum_{i \in I} V_i(\hat{g}_0(X_i) - g_0(X_i)) + \dots$$

To let c^* approach 0, we split the N sample into two $N/2$ samples.

Auxiliary Sample (I^c)
estimates $\hat{g}_0, \hat{m}_0$

Main Sample (I)
estimates $\check{\theta}_0$

$$\hat{V} = D - \hat{m}_0(X)$$


Empirical Moment Formula

$$\psi(W; \theta, \eta) := \{Y - l(X) - \theta(D - m(X))\}(D - m(X)) \text{ with } \eta = (l, m)$$

$$l_0(X) = E[Y|X] = m_0(X)\theta_0 + g_0(X)$$

Mathematical Deduction

$$l_0(X) = m_0(X)\theta_0 + g_0(X) \implies \mathbf{g_0(X) = l_0(X) - \theta_0 m_0(X)}$$

$$\{Y - D\theta - g(X)\} \implies \{Y - D\theta - (l(X) - \theta m(X))\} \implies \{Y - D\theta - l(X) + \theta m(X)\}$$

Collecting θ , we obtain:

$$\{Y - l(X) - \theta(D - m(X))\}$$

Road Map DML: Steps 1 & 2

1. Cross-fitting (Data Splitting)

- Divide the total N observations into K folds.
- For each fold k :
 - Use I_k^c (**Auxiliary Sample**) to train the ML algorithms.
 - Use I_k (**Main Sample**) to perform inference.



2. Nuisance Parameter Estimation

On the **Auxiliary Sample** (I_k^c), we estimate the mapping between controls X and variables Y, D using any ML tool:

$$\hat{l}_k(X) = E[Y|X] \quad \text{and} \quad \hat{m}_k(X) = E[D|X]$$

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Road Map DML: Steps 3 & 4

3. Partialling Out (Residualization)

On the **Main Sample** (I_k), we remove the "noise" explained by the controls X to obtain **orthogonal residuals**:

$$\widehat{W} = Y - \widehat{l}_k(X) \quad \text{and} \quad \widehat{V} = D - \widehat{m}_k(X)$$



4. Solving the Neyman-Orthogonal Score

We estimate $\widehat{\theta}_k$ on the residuals of I_k . The final DML estimator is the average across all K folds:

$$\widehat{\theta}_{DML} = \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{n} \sum_{i \in I_k} \widehat{V}_i D_i \right)^{-1} \frac{1}{n} \sum_{i \in I_k} \widehat{V}_i \widehat{W}_i$$

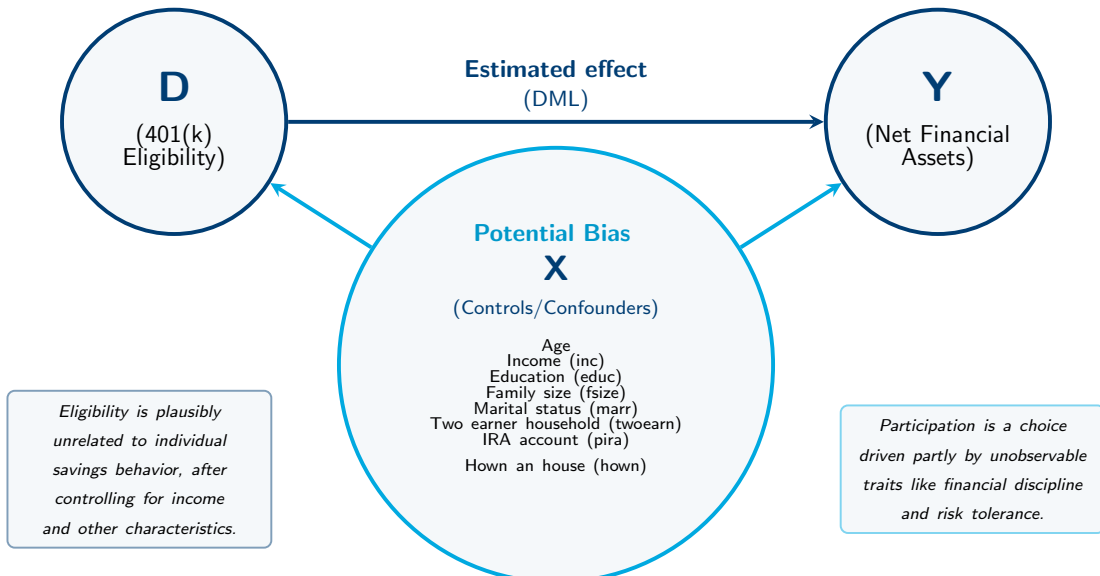
Result: $\sqrt{N}$ consistency even with biased ML nuisance estimators.

Introduction to 401(K) Dataset

The sample is drawn from the **1991 Survey of Income and Program Participation (SIPP)** and consists of 9,915 observations. The observational units are household reference persons aged 25-64 and spouse if present. Households are included in the sample if at least one person is employed and no one is self-employed.

	Y		D	X								Others		
Stat	net_tfa	e401	age	inc	educ	fsize	marr	twoearn	pira	hown	db	nifa	tw	p401
Mean	18,051.5	0.37	41.1	37,200.6	13.2	2.9	0.60	0.38	0.24	0.64	0.27	13,928.6	63,816.8	0.26
Std	63,522.5	0.48	10.3	24,774.3	2.8	1.5	0.49	0.49	0.43	0.48	0.44	54,904.9	111,529.7	0.44
Min	-502,302.0	0.00	25.0	-2,652.0	1.0	1.0	0.00	0.00	0.00	0.00	0.00	0.0	-502,302.0	0.00
25%	-500.0	0.00	32.0	19,413.0	12.0	2.0	0.00	0.00	0.00	0.00	0.00	200.0	3,291.5	0.00
50%	1,499.0	0.00	40.0	31,476.0	12.0	3.0	1.00	0.00	0.00	1.00	0.00	1,635.0	25,100.0	0.00
75%	16,524.5	1.00	48.0	48,583.5	16.0	4.0	1.00	1.00	0.00	1.00	1.00	8,765.5	81,487.5	1.00
Max	1,536,798.0	1.00	64.0	242,124.0	18.0	13.0	1.00	1.00	1.00	1.00	1.00	1,430,298.0	2,029,910.0	1.00

Causal Inference Structure



Neyman Orthogonal Scores for Various Target Parameters

Observed Variables (W)	Nuisance parameters (η)	Neyman orthogonal score ($m(W; \theta, \eta)$)
(i) Target Parameter: <i>Linear regression coefficient</i>		
Y: outcome	$\eta_{Y,0} = \arg \min_{\eta} E[(Y - X'\eta)^2]$	$[(Y - X'\eta_Y) - \theta(D - X'\eta_D)](D - X'\eta_D)$
D: treatment	$\eta_{D,0} = \arg \min_{\eta} E[(D - X'\eta)^2]$	
X: controls		
(ii) Target Parameter: <i>Partially linear regression coefficient</i>		
Y: outcome	$\ell_0(X) = E[Y \mid X]$	$[(Y - \ell(X)) - \theta(D - r(X))](D - r(X))$
D: treatment	$r_0(X) = E[D \mid X]$	
X: controls		
(iii) Target Parameter: <i>Linear IV coefficient (No controls)</i>		
Y: outcome		$(Y - \theta D)r(Z)$
D: treatment	$r_0(Z) = E[D \mid Z]$	
Z: instruments		

Observed Variables (W)	Nuisance parameters (η)	Neyman orthogonal score ($m(W; \theta, \eta)$)
(iv) Target Parameter: <i>Average treatment effect</i> ($E[E[Y \mid D = 1, X] - E[Y \mid D = 0, X]]$)		
Y: outcome	$\ell_0(D, X) = E[Y \mid D, X]$	$\alpha(D, X)(Y - \ell(D, X)) + \ell(1, X) - \ell(0, X) - \theta$
D: binary treatment	$\alpha_0(D, X) = \frac{D}{r_0(X)} - \frac{1-D}{1-r_0(X)}$	
X: controls	for $r_0(X) = E[D \mid X]$	
(v) Target Parameter: <i>Partially linear regression coeff. with excluded instruments</i>		
Y: outcome	$\ell_0(X) = E[Y \mid X]$	$[(Y - \ell(X)) - \theta(D - r(X))](Z - h(X))$
D: treatment	$r_0(X) = E[D \mid X]$	
Z: instrument	$h_0(X) = E[Z \mid X]$	
X: controls		
(vi) Target Parameter: <i>Average structural derivative</i> ($E[\frac{\partial}{\partial d} E[Y \mid D = d, X]]_{d=D}$)		
Y: outcome	$\ell_0(D, X) = E[Y \mid D, X]$	$\frac{\partial}{\partial d} \ell_0(d, X) _{d=D} + \alpha(D, X)(Y - \ell(D, X)) - \theta$
D: continuous treat.	$\alpha_0(D, X) = -\frac{\partial}{\partial d} \log f_0(d, X) _{d=D}$	
X: controls		